Analysis and Synthesis Prior Greedy Algorithms for Non-linear Sparse Recovery

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In this work we address the problem of recovering sparse solutions to non-linear inverse problems. We look at two variants of the basic problem – the synthesis prior problem when the solution is sparse and the analysis prior problem where the solution is co-sparse in some linear basis. For the first problem, we propose non-linear variants of the Orthogonal Matching Pursuit (OMP) and CoSamp algorithms; for the second problem we propose a non-linear variant of the Greedy Analysis Pursuit (GAP) algorithm. We empirically test the success rates of our algorithms on exponential and logarithmic functions.

The experiments are evaluated based on empirical success rates. For each configuration, 1000 trials are generated; a trial is considered successful if the normalized mean squared error is below 10^{-3}.

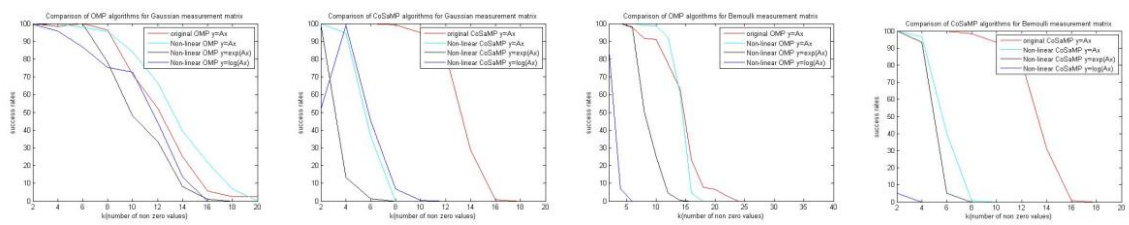


Figure 1. Comparison of success rates for various functions with OMP and CoSaMP for i.i.d. Gaussian and Bernoulli measurement matrix.

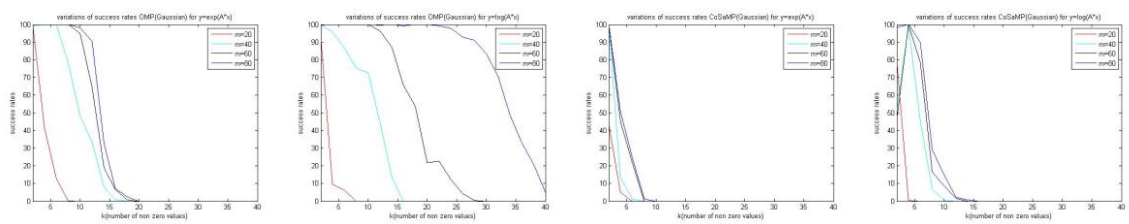
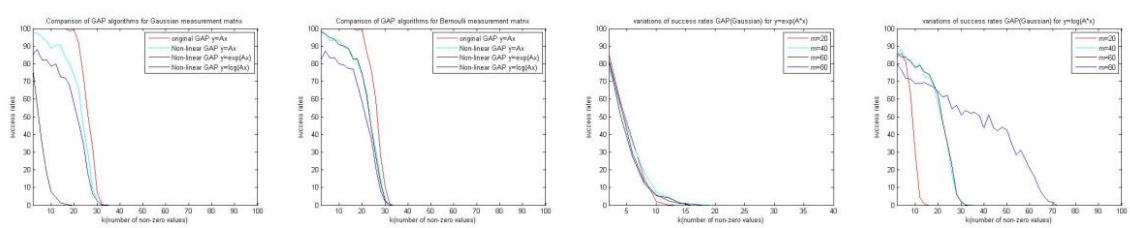


Figure 2. Variation of success rates for OMP and CoSaMP with number of non-zero values and number of measurements for i.i.d Gaussian matrix.



(a) (b) (c) (d)

Figure 3. (a) and (b) Comparison of success rates for various functions with GAP for i.i.d Gaussian and Bernoulli measurement matrix. (c) and (d) Variation of success rates for GAP with number of non-zero values and number of measurements for i.i.d Gaussian matrix.

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